

Continuity property of probability

$a_1, a_2, a_3, \dots$

$b_1, b_2, b_3, \dots$

$$\lim_{n \rightarrow \infty} a_n = a^*$$

$$\lim_{n \rightarrow \infty} b_n = b^*$$

$$\lim_{n \rightarrow \infty} (a_n + b_n) = a^* + b^*$$

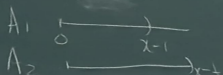
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right)$$

Corollary 1: If $A_1 \subseteq A_2 \subseteq A_3 \dots$

then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_n\right) \\ = \lim_{n \rightarrow \infty} P(A_n)$$

Example: $A_n = \left[0, x - \frac{1}{n}\right)$ for some $x < 1$



$$A_1 \subseteq A_2 \subseteq \dots$$

$$P(A_n) = \left(x - \frac{1}{n}\right)$$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n) \\ = \lim_{n \rightarrow \infty} \left(x - \frac{1}{n}\right) \\ = x$$

Corollary 2: If $A_1 \supseteq A_2 \supseteq A_3 \dots$

$$\text{then } P\left(\bigcap_{i=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n)$$

$$\Rightarrow A_1^c \subseteq A_2^c \subseteq \dots$$

From Corollary 1: $P\left(\bigcup_{i=1}^{\infty} A_i^c\right) = \lim_{n \rightarrow \infty} P(A_n^c)$

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1 - P\left(\bigcup_{i=1}^{\infty} A_i^c\right) \\ = 1 - P\left(\bigcup_{i=1}^{\infty} A_i^c\right) \text{ (De Morgan's law)} \\ = 1 - \lim_{n \rightarrow \infty} P(A_n^c) \\ = 1 - \lim_{n \rightarrow \infty} (1 - P(A_n)) \\ = 1 - 1 + \lim_{n \rightarrow \infty} P(A_n)$$

Union bound

$$P\left(\bigcup_{i=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n)$$

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$$\lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right)$$

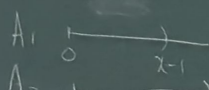
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Example: $A_n = \left[0, x - \frac{1}{n}\right)$ for some $x < 1$



$$A_1 \subseteq A_2 \subseteq \dots$$

$$P(A_n) = \left(x - \frac{1}{n}\right)$$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n) \\ = \lim_{n \rightarrow \infty} \left(x - \frac{1}{n}\right) \\ = x$$

Corollary 2: If $A_1 \supseteq A_2 \supseteq A_3 \dots$

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Conditional probability

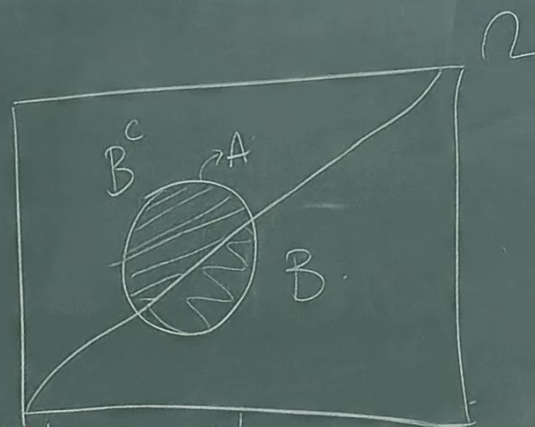
① Thank x

What is the probability that second word is you or her or him given that first word is Thank

② You designed a system to detect enemy aeroplanes. The system says it (RADAR) detected an aeroplane. What is the probability (false positive) there was actually an aeroplane.

B : event you know has happened.

$$P(A|B) \propto P(A \cap B)$$



Intuitively we want

$$P(B|B) = 1$$

Definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) > 0$$

Does conditional probability satisfy

the probability axioms? Given some event B ,

① Non-negativity:

$$P(B) > 0$$
$$\boxed{P(A|B)} = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) \geq 0 \quad \forall A \in \mathcal{F}$$

② Normalization

$$P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)}$$
$$= \frac{P(B)}{P(B)} = 1$$

③ Additivity

$A_1, A_2, \dots$ are pairwise disjoint

$$P(A \cap B) = P(A|B) P(B)$$

$$\text{Multiplicative rule} \quad = P(B|A) P(A)$$

Similarly

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \\ \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right)$$

$$= \frac{P\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right)}{P(B)}$$

distributive
property

$$= \frac{P\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{P(B)}$$

$$P(B)$$

$(A_i \cap B), (A_j \cap B)$ are

disjoint

additivity
axiom

as $A_i \cap A_j = \emptyset$ for $i \neq j$

$$= \frac{\sum_{i=1}^{\infty} P(A_i \cap B)}{P(B)}$$

$$P(B)$$

Consider an experiment where you pick ^{three} cards from a deck of 52 cards without replacement.

What is the probability that all the cards are hearts.

A_i = event that i -th card is a heart.

13 hearts

13 diamonds

13 spade

13 clubs.

$$P(A_1) = \frac{13}{52} = \frac{1}{4}$$

$$P(A_2 | A_1) = \frac{12}{51}$$

$$P(A_3 | A_1 \cap A_2) = \frac{11}{50}$$

→ 12 hearts
13 dia
13
13

$$P(A_1 \cap A_2 \cap A_3) = \frac{1}{4} \times \frac{12}{51} \times \frac{11}{50} = \frac{11}{850}$$